Formulas

AQA A-LEVEL BUSINESS 7132 - ALL UNITS

All the formulas on AQA's 7131/7132 formula list (revised August 2018), plus the calculation methods the specification requires that are not on that list. These are not given to you in the exam, so they need learning. Employee retention rate is no longer on the specification and is left out. Always show your working, and include £, % or units in your answer.

Unit 3.1 What is business?

Measure	Formula	Example	Watch out
Revenue	Selling price per unit × units sold	£3.00 × 4,000 = £12,000	Revenue is also called sales or turnover. It is not profit.
Total variable costs	Variable cost per unit × units sold	£0.80 × 4,000 = £3,200	
Total costs	Fixed costs + variable costs	£6,000 + £3,200 = £9,200	
Profit	Total revenue - total costs or total contribution - fixed costs	£12,000 - £9,200 = £2,800	A negative answer is a loss.
Market capitalisation	Number of issued shares × current share price	20m × £2.50 = £50m	Not the value of the company's assets.

Unit 3.2 Managers, leadership and decision making

Measure	Formula	Example	Watch out
Expected value	Sum of (probability × pay-off) for each outcome	(0.6 × £800,000) + (0.4 × £100,000) = £520,000	Probabilities at each chance node add up to 1.
Net gain	Expected value - initial cost of the decision	£520,000 - £200,000 = £320,000	Not total pay-off minus cost.

Unit 3.3 Marketing management

Measure	Formula	Example	Watch out
Market growth (%)	Change in market size ÷ original market size × 100	£6m ÷ £40m × 100 = 15%	Divide by the original (earlier) figure.
Sales growth (%)	Change in sales ÷ original sales × 100	£0.52m ÷ £5m × 100 = 10.4%	From the spec - not on AQA's formula list.
Market share (%)	Sales of one product, brand or business ÷ total sales of the market × 100	£5.52m ÷ £46m × 100 = 12%	Check whether data is by value (£) or volume (units).

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Unit 3.3 Marketing management

Measure	Formula	Example	Watch out
Market size	Total sales of all businesses in the market	Sales of one firm ÷ its market share: £5.52m ÷ 0.12 = £46m	From the spec - not on AQA's formula list.
Price and income elasticity	Interpret only - you do not need to calculate elasticity	PED -0.4, price +10% → quantity -4%	Inferior goods have negative YED.

Unit 3.4 Operational management

Measure	Formula	Example	Watch out
Added value	Sales revenue - costs of bought-in goods and services	£50,000 - £18,000 = £32,000	Not the same as profit.
Labour productivity	Output over a time period ÷ number of employees	1,500 ÷ 25 = 60 units per employee per week	Give units and the time period.
Unit cost (average cost)	Total costs ÷ number of units of output	£30,000 ÷ 1,500 = £20	
Capacity utilisation (%)	Actual output ÷ maximum possible output × 100	1,500 ÷ 2,000 × 100 = 75%	100% is not necessarily ideal.

Unit 3.5 Financial management

Measure	Formula	Example	Watch out
Return on investment (%)	Profit from investment ÷ cost of investment × 100	£30,000 ÷ £200,000 × 100 = 15%	
Gross profit	Revenue - cost of sales	£500,000 - £300,000 = £200,000	
Operating profit	Gross profit - operating expenses	£200,000 - £120,000 = £80,000	Also called profit from operations.
Profit for the year	Operating profit + profit from other activities - net finance costs - tax	£80,000 - £10,000 - £14,000 = £56,000	
Gross profit margin (%)	Gross profit ÷ revenue × 100	£200,000 ÷ £500,000 × 100 = 40%	A margin is a %; a profit is £.
Operating profit margin (%)	Operating profit ÷ revenue × 100	£80,000 ÷ £500,000 × 100 = 16%	
Profit for the year margin (%)	Profit for the year ÷ revenue × 100	£56,000 ÷ £500,000 × 100 = 11.2%	

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Unit 3.5 Financial management

Measure	Formula	Example	Watch out
Variance	Budgeted figure - actual figure	Revenue: £50,000 - £46,000 = £4,000 adverse	Favourable or adverse depends on the effect on profit, not the sign.
Net cash flow and closing balance	Net cash flow = cash inflows - cash outflows Closing balance = opening balance + net cash flow	£20,000 - £27,000 = -£7,000; £5,000 - £7,000 = -£2,000	From the spec - not on AQA's formula list.
Contribution per unit	Selling price - variable cost per unit	£25 - £10 = £15	
Total contribution	Contribution per unit × units sold or total revenue - total variable costs	£15 × 4,200 = £63,000	Total revenue is not total costs + total contribution.
Break-even output	Fixed costs ÷ contribution per unit	£45,000 ÷ £15 = 3,000 units	Answer in units.
Margin of safety	Actual output - break-even output	4,200 - 3,000 = 1,200 units	In units, not £. Not the same as profit.

Unit 3.6 Human resource management

Measure	Formula	Example	Watch out
Labour turnover (%)	Number of staff leaving ÷ number of staff employed × 100	12 ÷ 80 × 100 = 15%	
Labour productivity	Output over a time period ÷ number of employees	90,000 ÷ 80 = 1,125 units per employee	
Employee costs as % of turnover	Employee costs ÷ turnover × 100	£360,000 ÷ £1.2m × 100 = 30%	Turnover here means revenue.
Labour cost per unit	Labour costs ÷ units of output	£360,000 ÷ 90,000 = £4	

Unit 3.7 Analysing the strategic position of a business

Measure	Formula	Example	Watch out
Return on capital employed (%)	Operating profit ÷ (total equity + non-current liabilities) × 100	£2m ÷ £10m × 100 = 20%	Capital employed = total equity + non-current liabilities.
Current ratio	Current assets ÷ current liabilities	£600,000 ÷ £400,000 = 1.5:1	Express as x:1.

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Unit 3.7 Analysing the strategic position of a business

Measure	Formula	Example	Watch out
Gearing (%)	Non-current liabilities ÷ (total equity + non-current liabilities) × 100	£2m ÷ £10m × 100 = 20%	
Payables days	Payables ÷ cost of sales × 365	£90,000 ÷ £1.2m × 365 = 27.4 days	Uses cost of sales.
Receivables days	Receivables ÷ revenue × 365	£150,000 ÷ £1.825m × 365 = 30 days	Uses revenue.
Inventory turnover	Cost of sales ÷ average inventories held	£1.2m ÷ £100,000 = 12 times a year	A number of times, not days.
Payback	Years until cumulative net cash flow covers the cost. Part year: amount still needed ÷ next year's net cash flow × 12	£30,000 ÷ £50,000 × 12 = 7.2 months → 2 years 7.2 months	Method from the spec - not on AQA's formula list.
Average rate of return (%)	Average annual return ÷ initial cost of project × 100 Average annual return = (total net cash flows - initial cost) ÷ years	(£160,000 - £100,000) ÷ 4 = £15,000; £15,000 ÷ £100,000 × 100 = 15%	Subtract the initial cost first.
Net present value	Net cash flow × discount factor = present value NPV = total present values - initial cost	£125,180 - £100,000 = £25,180	Discount factors are given. Method from the spec - not on AQA's formula list.

Unit 3.10 Managing strategic change

Measure	Formula	Example	Watch out
Total float	Latest finish time - duration - earliest start time	Activity C: 7 - 2 - 3 = 2 days	Method from the spec - not on AQA's formula list. Critical activities have zero float.